PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Acoustic Signal-Based Non-Intrusive Engine Fault Detection Using Mel-Spectrograms and Convolutional Neural Networks

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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INDEX

DECLARATION

I hereby certify that the work which is being presented in the project report entitled " Acoustic Signal-Based Non-Intrusive Engine Fault Detection Using Mel-Spectrograms and Convolutional Neural Networks " in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Gurpreet Singh. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place:Rajpura **Harsimrat Singh**
Date: April 28, 2026 (2210990395)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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List Of Figures and Tables

List of Figures

List of Tables

Abstract

Engine faults can reduce vehicle performance, increase fuel consumption, create vibrations, and lead to expensive repairs if ignored for a long time. Common problems such as misfire, knocking, rough idling, ticking noise, and belt squeal usually begin with small changes in engine sound. Traditional diagnosis methods often depend on workshop inspection, mechanic experience, or tools such as OBD scanners, which may not always be easily available to normal users.

This project presents an acoustic signal-based non-intrusive method for engine fault detection using machine learning. Instead of opening engine parts or connecting diagnostic equipment, the system analyses recorded engine sound signals. The audio data is first segmented and preprocessed, then converted into **Mel-spectrogram** images that represent frequency changes over time. These spectrograms are used as input to a **Convolutional Neural Network (CNN)** model for classification.

The model was trained using the **AI Mechanic Dataset** along with **11 additional real-world engine recordings** collected from publicly available sources. Audio samples were grouped into five classes: Normal Engine, Combustion Fault, Mechanical Fault, Belt/Accessory Fault, and Background Noise. The final model achieved approximately **86% multi-class classification accuracy**. Results also showed strong class-wise performance through confusion matrix and ROC curve analysis.

The project demonstrates that engine sound contains useful diagnostic information and can be used for low-cost early fault detection. With further improvements, such systems can be useful for vehicle owners, garages, and predictive maintenance applications.

1. Introduction

Engines are one of the most important parts of any vehicle, and their condition directly affects power, mileage, emissions, and overall reliability. Even small engine faults can become serious if they are not detected early. Problems such as misfire, spark plug failure, knocking, worn belts, rough idling, and mechanical wear usually change the normal sound pattern of the engine before major damage happens.

Many experienced mechanics can often identify issues by listening carefully to the engine sound. A healthy engine usually produces a balanced and regular sound, while faults create irregular noises such as metallic knocking, uneven firing pulses, squealing belts, ticking sounds, or abnormal vibration noise. This shows that engine sound can be an important source of information about engine health.

Most modern vehicles have onboard diagnostic systems, but these often require scanners, technical knowledge, or workshop visits. Many normal users may ignore early warning signs because they do not have tools available or are unsure about the actual problem.

With recent growth in artificial intelligence and deep learning, it has become possible to train systems that automatically learn patterns from sound data. Audio-based AI is already used in speech recognition, security systems, and environmental sound classification. This project applies the same concept to engine fault diagnosis.

The main idea of this work is to use engine sound recordings, convert them into **Mel-spectrogram images**, and train a **CNN model** to classify different engine conditions. Mel-spectrograms are useful because they show how sound frequency changes over time in a visual format. CNN models are suitable because they are effective in image pattern recognition and can learn fault-related sound features automatically.

The proposed system was trained on five engine sound categories and achieved approximately **86% accuracy**, showing that sound-based fault detection is practical even without expensive tools or contact sensors.

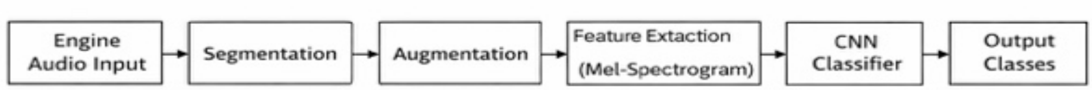

Fig. 1. Proposed framework for engine fault detection system.

Fig. 1. Proposed workflow of the system

2. Problem Statement and Objectives

Engine faults often begin as minor issues such as rough idling, knocking sounds, irregular combustion, worn belts, ticking noise, or vibration. In many cases, vehicles continue to run, so users delay inspection and continue driving. However, if these faults are ignored, they can lead to reduced engine performance, poor fuel efficiency, higher emissions, breakdowns, and costly repairs.

The common methods used for engine diagnosis include manual inspection by mechanics, workshop testing, or electronic diagnostic tools such as OBD-II scanners. These methods are useful, but they also have some limitations. They require technical knowledge, proper tools, and access to a service center. This may not always be convenient for normal users.

At the same time, most engine problems produce noticeable changes in sound. Misfire can create uneven firing pulses, knocking creates sharp metallic sounds, worn belts create squealing noise, and mechanical wear may create ticking or rattling sounds. If these sound differences can be analysed using machine learning, then engine faults may be detected in a simple and non-intrusive way.

The main problem addressed in this project is to develop a system that can identify engine conditions using only recorded sound signals, without using extra sensors or opening engine components.

The primary objective of this project was to design and implement a machine learning model capable of classifying engine sounds into different operating and fault conditions.

The main objectives of the project are:

- To collect engine sound recordings from multiple sources and prepare a labelled dataset.
- To use the **AI Mechanic Dataset** and **11 additional real-world recordings** for better diversity.
- To group recordings into five classes: Normal, Combustion Fault, Mechanical Fault, Belt/Accessory Fault, and Background Noise.
- To apply preprocessing techniques such as resampling to **22,050 Hz**, mono conversion, segmentation, and normalization.
- To divide audio into fixed **3-second clips** for uniform model input.
- To apply augmentation techniques such as time shifting, Gaussian noise addition, and pitch variation.
- To convert audio clips into **128 × 128 Mel-spectrogram** images.
- To train a **Convolutional Neural Network (CNN)** model using the generated spectrograms.
- To evaluate the model using accuracy, confusion matrix, ROC curves, precision, recall, and F1-score.
- To achieve practical engine fault classification performance (around **86% accuracy**).

3. Background

Condition monitoring using sound signals has been used in many industries for a long time. Machines such as motors, pumps, compressors, generators, turbines, and engines usually produce regular sound patterns during normal operation. When wear, looseness, imbalance, or internal faults begin to develop, these sound patterns start changing. Because of this, sound analysis is considered a useful non-contact method for detecting abnormal conditions.

In the automotive field, mechanics often use sound as an early indicator of engine condition. A smooth engine idle usually has a balanced and repeating sound, while faults can create irregular noises. Misfire may create uneven combustion pulses, knocking may create sharp metallic impacts, worn belts may create squealing noise, and internal wear may create ticking or rattling sounds. This shows that engine sound contains meaningful diagnostic information.

Directly analysing raw audio waveforms can be difficult because recordings may have different volume levels, different microphones, background noise, and different operating conditions. To make the data easier to analyse, sound can be converted into a **spectrogram**, which is a time-frequency representation showing how sound frequencies change over time.

For this project, **Mel-spectrograms** were used. A Mel-spectrogram uses the Mel scale, which is closer to how humans naturally hear sound. It helps highlight useful sound patterns while reducing less important frequency detail. Mel-spectrograms are widely used in speech recognition, music analysis, and environmental sound classification.

Recent developments in deep learning have made audio classification more effective. **Convolutional Neural Networks (CNNs)**, originally designed for image recognition, can also work well on spectrogram images. CNN models automatically learn patterns such as edges, shapes, textures, and repeating structures without needing manual feature engineering.

Since a Mel-spectrogram converts engine sound into an image-like format, CNN models become a strong choice for engine fault detection. By combining Mel-spectrogram feature extraction with CNN classification, the system can learn the difference between healthy and faulty engine sounds in a practical way.

4. Methodology

4.1 Dataset Collection and Preparation

For this project, engine sound data was collected from two different sources to create a more useful and diverse dataset for training and testing the model.

The main source of data was the **AI Mechanic Dataset**, a publicly available labelled dataset containing recordings of multiple engine operating conditions and fault-related sounds. This dataset was selected because it provided organized training samples suitable for supervised learning.

To improve real-world relevance, a second source of data was also included. This consisted of **11 additional engine sound recordings** collected from publicly available online sources. These recordings were useful because they introduced natural variations such as:

- different vehicle types
- different microphone quality
- different recording distances
- surrounding environmental noise
- different engine operating conditions

Including these extra recordings helped reduce overdependence on a single controlled dataset and improved the diversity of the final training data.

Since some original fault labels were too specific or unevenly distributed, the recordings were grouped into five broader classes to make classification more practical and balanced.

Final Dataset Classes

1. **Normal Engine Operation**
   Smooth idling or normal running condition.
2. **Combustion Faults**
   Misfire, spark plug related issues, uneven firing sounds.
3. **Mechanical Faults**
   Knocking, ticking, rattling, internal wear sounds.
4. **Belt / Accessory Faults**
   Belt squeal, pulley noise, accessory drive sounds.
5. **Background Noise**
   Non-engine sounds such as traffic, wind, or surrounding environmental noise.

This grouping reduced unnecessary fragmentation of classes and made the final model easier to train and more practical for real-world engine fault detection.

The dataset was later segmented into fixed **3-second clips** and used for preprocessing, augmentation, feature extraction, and CNN training.

Table I. Typical Engine Faults and Their Acoustic Characteristics

Fault Type	Description	Acoustic Pattern
Normal Operation	Smooth engine running	Stable and consistent sound
Misfire	Incomplete combustion	Irregular bursts, uneven rhythm
Knocking	Premature combustion	Sharp metallic knocking sound
Belt / Accessory Fault	Loose or worn belt	Squealing or chirping sound
Mechanical Fault	Internal component wear	Repetitive ticking or knocking

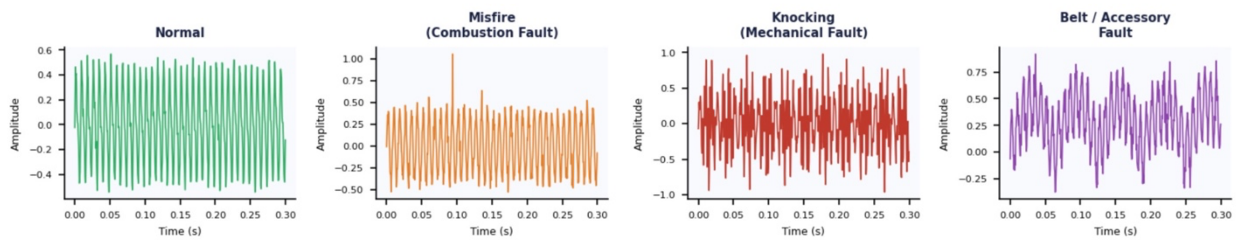

Fig.2. Simulated acoustic waveforms illustrating the characteristic time-domain differences between engine conditions.

4.2. Audio Preprocessing and Feature Extraction

The collected engine audio recordings were not identical in length, loudness, recording quality, or background environment. Because of this, preprocessing was necessary before training the model. Standardizing the data helps the neural network focus on actual fault-related sound patterns instead of random recording differences.

All audio files were first converted to a common sampling rate of **22,050 Hz** and changed to **mono channel** format. This reduced unnecessary variation and ensured that every recording followed the same input format.

After standardization, each recording was divided into fixed **3-second audio segments**. Using equal-length clips allowed the model to receive a consistent input size during training. If the final segment of any recording was shorter than the required duration, **zero-padding** was applied so that the clip still matched the expected length.

To improve dataset diversity and make the model more robust, augmentation techniques were also used. These methods helped simulate real-world recording variations and reduced overfitting. The following augmentation methods were applied:

- **Time Shifting** – small forward or backward movement of the waveform.

- **Gaussian Noise Addition** – adding low random noise to simulate environmental conditions.
- **Pitch Variation** – slight pitch change to represent engine speed variation.

After preprocessing, each audio clip was converted into a **Mel-spectrogram**. A Mel-spectrogram is a visual representation of sound that shows how frequency content changes over time using the Mel scale, which is closer to human hearing behaviour.

This method was chosen because engine faults often create frequency-based changes such as impulses, irregular harmonics, rattling energy, or tonal belt noise. These patterns become easier to detect in Mel-spectrogram form than in raw waveform form.

The generated spectrograms were resized to **128 × 128 pixels** and normalized before being used as input to the CNN model.

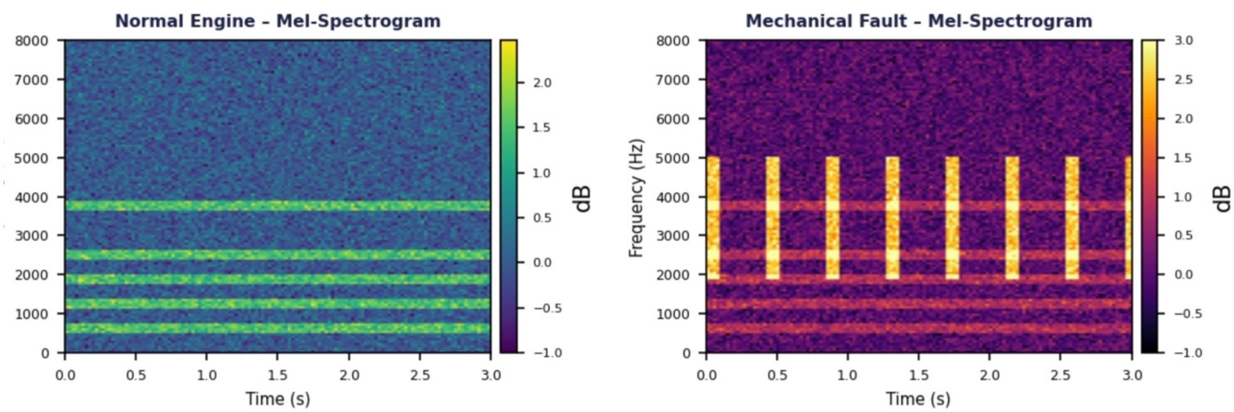

Fig. 3. Example Mel-spectrograms of Normal Engine and Faulty Engine conditions.

4.3. Proposed CNN Model

A **Convolutional Neural Network (CNN)** was selected for this project because CNNs are highly effective at recognising patterns in image data. Since Mel-spectrograms convert audio signals into image-like representations, CNN models are well suited for learning fault-related sound features automatically.

The proposed network used a simple and efficient architecture suitable for the available dataset size. The first convolution layer contained **32 filters** with a **3 × 3 kernel** and ReLU activation. This layer helped detect basic local features such as edges and frequency patterns in the spectrogram.

It was followed by a **max-pooling layer**, which reduced feature dimensions while preserving important information.

A second convolution layer with **64 filters** was then used to learn deeper and more complex sound patterns. Another max-pooling layer followed this stage.

After feature extraction, the output was passed through a **flatten layer**, which converted the feature maps into a one-dimensional vector.

This vector was connected to a **dense fully connected layer with 128 neurons**, allowing the network to learn higher-level classification relationships.

A **dropout layer** was added to reduce overfitting by randomly disabling some neurons during training.

The final output layer used **Softmax activation** to classify each sample into one of the five engine sound categories.

Training Parameters

- Optimizer: **Adam**
- Loss Function: **Sparse Categorical Crossentropy**
- Batch Size: **32**
- Epochs: **12**
- Input Size: **128 × 128 × 1**

For evaluation, the dataset was divided into **80% training data** and **20% testing data**. In addition, **5-fold cross-validation** was applied to obtain more reliable and consistent results.

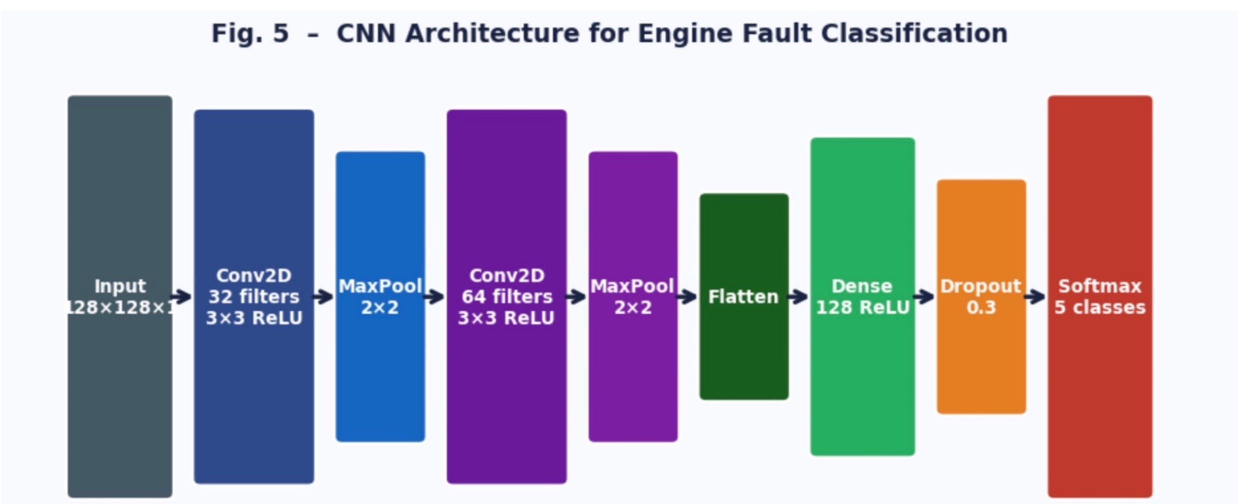

Fig. 4. Proposed CNN Architecture for Engine Fault Classification.

Table 2 CNN Layers and Output Configuration.

Layer	Type	Details
1	Conv2D	32 filters, 3×3
2	MaxPooling	2×2
3	Conv2D	64 filters, 3×3
4	MaxPooling	2×2
5	Flatten	-
6	Dense	128 units
7	Dropout	0.3
8	Output	5 classes

5.Tools and Technologies

The following tools and technologies were used in this project:

Programming Language:
Python

Libraries and Frameworks:
TensorFlow / Keras – for building and training CNN model
Librosa – for audio processing and Mel-spectrogram generation
NumPy – for numerical operations
Matplotlib – for visualization
Scikit-learn – for evaluation metrics

Development Environment:
Google Collab / VS Code

Version Control:
GitHub for code management and submission

Hardware:
Standard laptop/PC with CPU-based training

6. Results

The trained CNN model produced encouraging results for multi-class engine fault classification. The final system achieved approximately **86% overall classification accuracy**, showing that the model successfully learned meaningful acoustic differences between normal and faulty engine conditions.

This result indicates that engine sound contains enough diagnostic information to identify operating conditions without using scanners or contact sensors.The training process showed stable learning behaviour. Training and validation accuracy gradually improved across epochs, while loss values decreased steadily. This suggests that the model learned effectively without severe overfitting.

Class-wise performance also showed useful results. **Normal engine sounds** were identified with high confidence because their sound patterns were more regular and consistent.

Mechanical fault classes showed strong recall, which is important because knocking, ticking, and abnormal internal sounds should be detected early in practical use.Some confusion was observed between **background noise** and certain noisy fault classes. This is expected because traffic noise, wind noise, or surrounding sounds may overlap with some engine frequency ranges.

ROC curve analysis also indicated strong class separation ability for most categories.The results support the idea that combining **Mel-spectrogram feature extraction** with **CNN classification** can provide an effective and low-cost method for engine fault diagnosis.

Even with a limited dataset and standard hardware, the model achieved practical performance, which shows good future potential for real-world use.

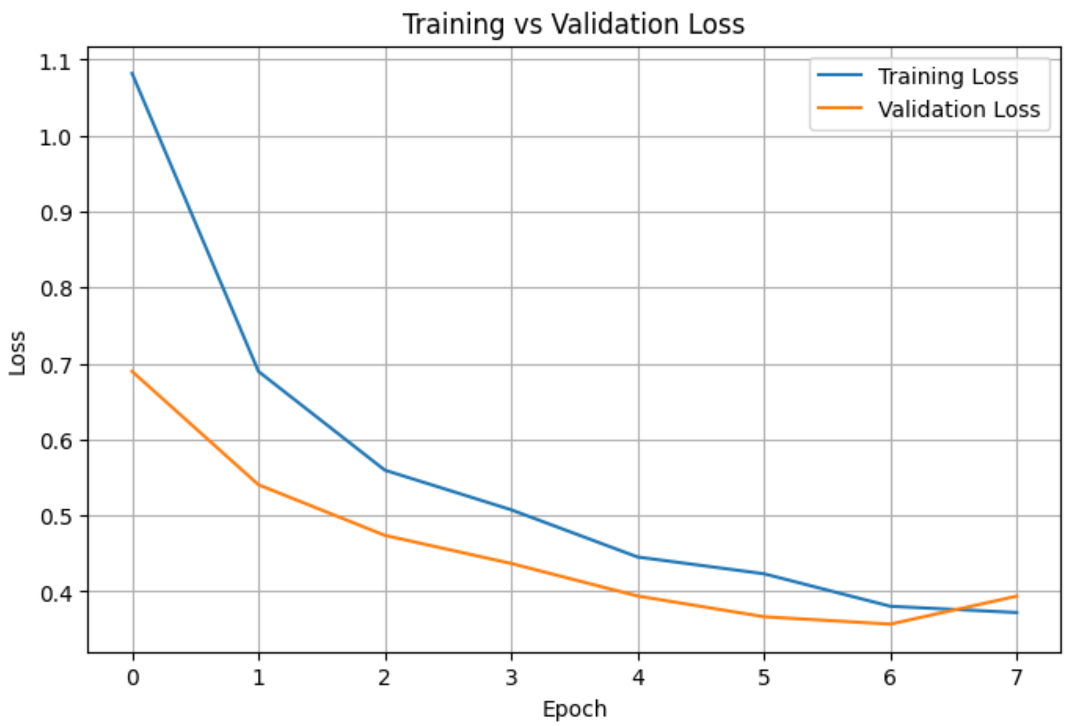

Fig. 5. Training and Validation Accuracy Graph.

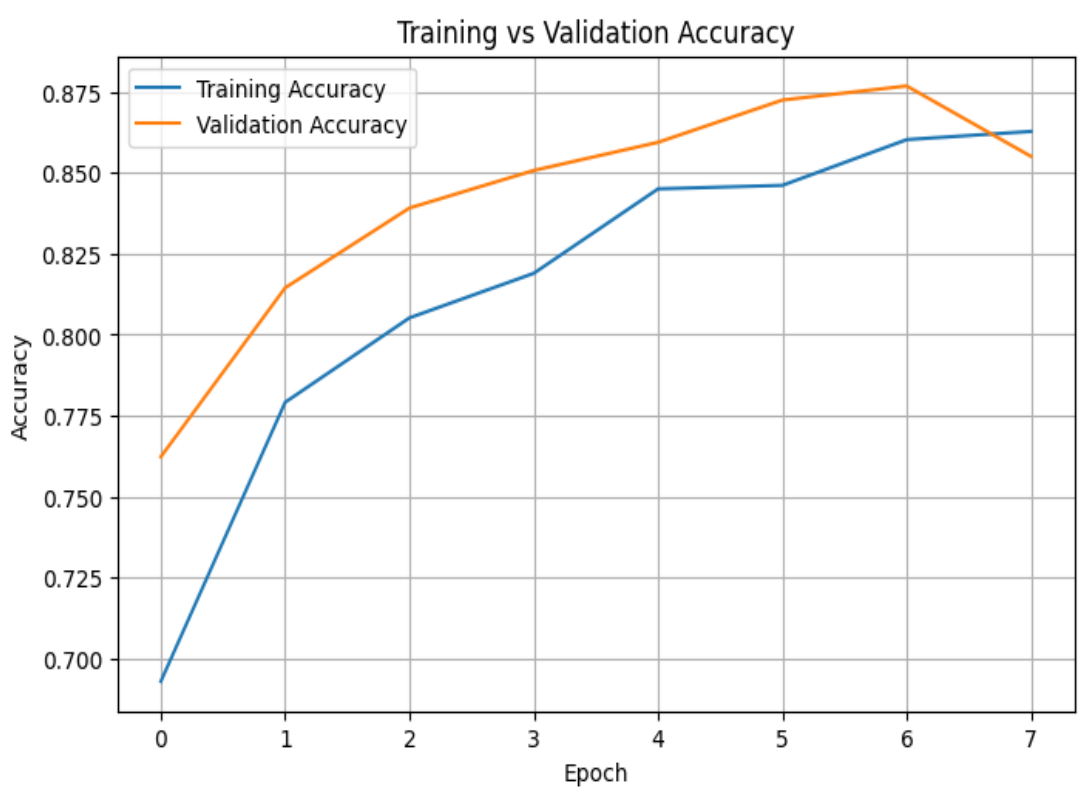

Fig. 6. Training and Validation Loss Graph.

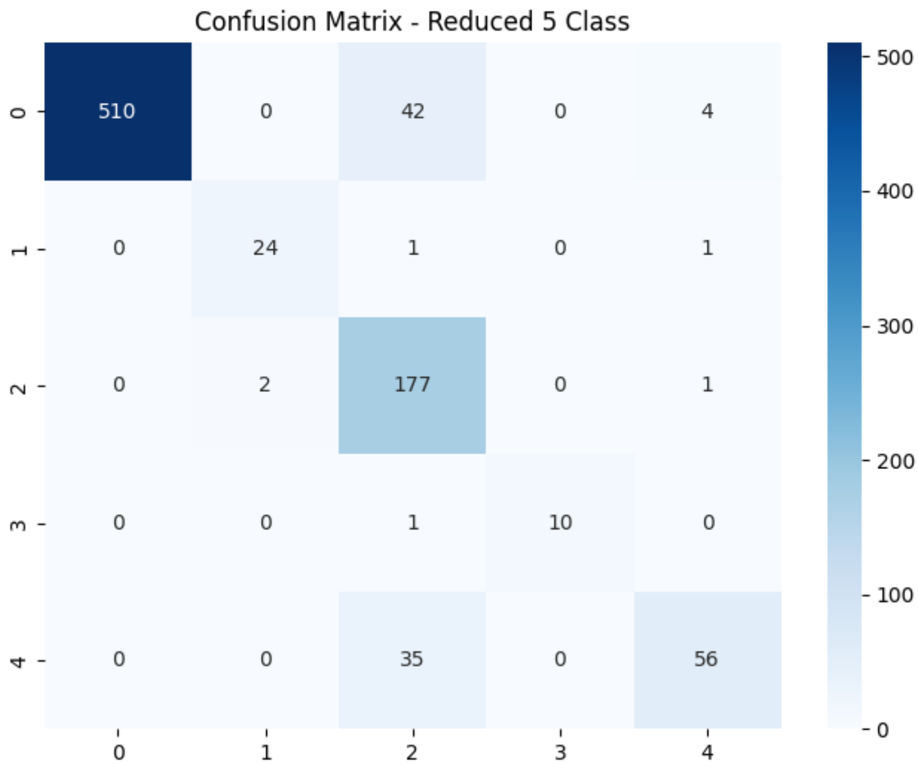

Fig. 7. Confusion Matrix for Five-Class Classification.

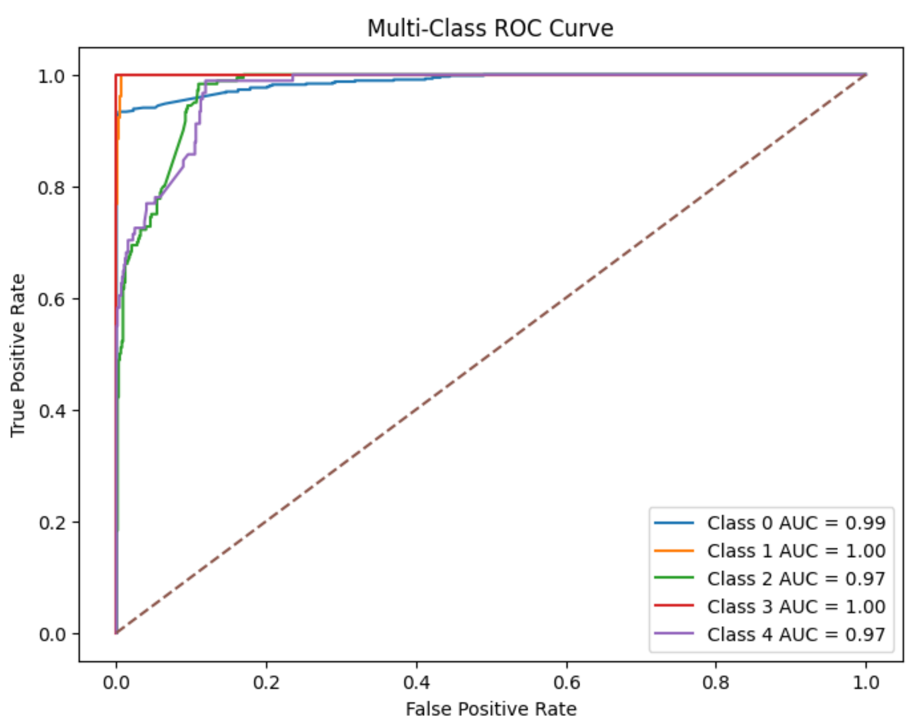

Fig. 8. ROC Curves for Multi-Class Engine Fault Detection.

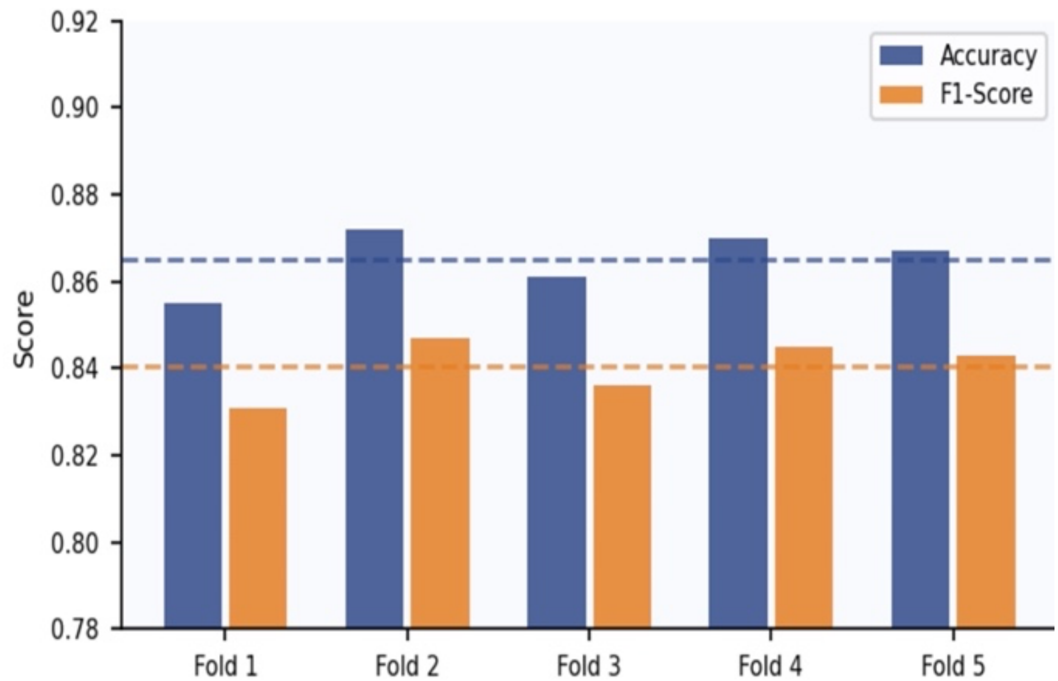

Fig. 9. Five fold Cross Validation Result

Table 3 Final Performance Metrics (Accuracy, Precision, Recall, F1-Score).

Metric	Value
Overall Accuracy	86.0%
Precision (Macro Avg.)	0.90
Recall (Macro Avg.)	0.87
F1-Score (Macro Avg.)	0.87
Precision (Weighted Avg.)	0.89
Recall (Weighted Avg.)	0.86
F1-Score (Weighted Avg.)	0.86

Class-wise Performance

Class	Precision	Recall	F1-Score
Normal Engine	1.00	0.92	0.96
Combustion Fault	0.92	0.92	0.92
Mechanical Fault	0.69	0.98	0.81
Belt / Accessory Fault	1.00	0.91	0.95
Background Noise	0.90	0.62	0.73

7. Improvements Made During Project Development

Several improvements were made during the development of the project to increase both technical quality and overall presentation. The initial version of the project focused mainly on basic model training, but later stages included better data preparation, stronger evaluation methods, cleaner code structure, and improved documentation.

The source code was reviewed and reorganized into a more structured format so that each stage of the pipeline could be understood and executed more easily. Separate sections were maintained for dataset loading, preprocessing, augmentation, feature extraction, model training, and result generation.

The dataset quality was also improved by applying augmentation techniques such as time shifting, pitch variation, and controlled noise addition. These methods increased data diversity and helped the model learn more robust sound patterns instead of depending only on limited original recordings.

To make the evaluation more reliable, **5-fold cross-validation** was added instead of depending only on a single train-test split. This provided a better idea of how consistently the model performs across different data partitions.

More detailed result outputs were also generated during later stages of the project. These included:

- Training and validation accuracy graphs
- Loss graphs
- Confusion matrix
- ROC curves
- Performance tables
- Mel-spectrogram sample outputs

These visual outputs made the model behaviour easier to analyse and explain.

The GitHub repository was also organized in a cleaner way with properly arranged source code, documentation, result files, and sample audio clips. This improved project presentation and made the work easier to review.

Overall, these improvements made the final version of the project more complete, more reliable, and stronger than the earlier drafts.

Link for Github Repository:
https://github.com/harsimrat11/Engine_Fault_detection_using_CNN

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8. Challenges and Limitations

Like most real-world machine learning projects, this work also faced several practical challenges and limitations.

One of the main difficulties was the limited availability of good quality publicly labelled engine sound datasets. Compared to areas such as image recognition or speech processing, engine fault audio datasets are smaller and less standardized. This restricts the amount of training data available.

Another challenge was **class imbalance**. Some fault categories had fewer samples than others, which can affect model learning and sometimes bias predictions toward classes with more examples.

Real-world audio recordings also create additional problems. Sounds captured in normal environments may include:

- Traffic noise
- Wind noise
- Human voices
- Microphone distortion
- Echo
- Different recording distances

These unwanted factors can reduce classification accuracy and make generalization harder.

Another limitation is that the current dataset does not fully represent every possible engine type or operating condition. For example:

- Rare W shaped engines
- Hybrid vehicles
- Motorcycles
- Heavily modified engines

were not fully covered in the present study.

Because of this, the current model should be considered a focused proof-of-concept system rather than a universal diagnostic solution for all vehicles.

The model also depends on audio quality. Very poor recordings or extremely noisy environments may affect prediction performance.

These limitations indicate that larger and more diverse datasets, along with more advanced models, would further improve future performance.

9. Conclusion

This project successfully demonstrated an **acoustic signal-based non-intrusive engine fault detection system** using Mel-spectrogram feature extraction and Convolutional Neural Networks.

The main idea of the work was that engine faults often create noticeable changes in sound patterns, and these acoustic differences can be analysed using machine learning techniques. Instead of requiring diagnostic scanners, external sensors, or physical engine inspection, the proposed model was able to classify engine operating conditions using only recorded sound signals.

This makes the approach simple, affordable, and convenient when compared with many traditional diagnosis methods that depend on workshop tools or mechanical inspection.

The final system achieved an overall accuracy of approximately **86%**, showing that meaningful fault information can be extracted directly from engine audio recordings. The obtained results also showed that **Mel-spectrograms** are effective for representing engine sound characteristics, while **CNN models** are capable of learning complex patterns linked with healthy and faulty engine behaviour.

Another important outcome of this project is that it demonstrates how artificial intelligence can be applied to practical automotive problems in an accessible way. Even with a limited dataset and a relatively simple CNN architecture, the model produced useful and encouraging results.

The study also showed that sound-based diagnostics can help in **early fault detection** before major mechanical damage occurs. Detecting issues such as misfire, knocking, belt noise, or abnormal engine operation at an early stage may help reduce repair costs, improve fuel efficiency, and increase vehicle reliability.

Such systems may be useful for:

- Individual vehicle owners
- Local workshops and garages
- Used-car inspection services
- Fleet operators
- Predictive maintenance systems

Although the current model is a proof-of-concept system, it provides a strong base for future development. With larger datasets, better model architectures, and real-time deployment, sound-based intelligent diagnostics can become an even more practical tool in the automotive industry.

Overall, the project achieved its objective of showing that engine sound is not just noise, but an important source of information that can be used for smart and non-intrusive fault diagnosis.

10. Future Scope

Although the current project produced encouraging results, there are several opportunities for future improvement and expansion.

One important future direction is **real-time engine fault detection** using a smartphone microphone or portable recording device. Instead of analysing only saved audio clips, the system could listen to live engine sound and provide instant feedback to the user.

Another major improvement would be the use of **larger and more diverse datasets**. Including recordings from different vehicle brands, engine sizes, ages, fuel types, and operating conditions would help the model generalize better and improve reliability.

The present work mainly focused on selected engine sound classes. Future versions can be extended to support:

- W shaped Engines
- Hybrid vehicles
- Motorcycles
- Generators
- Commercial vehicles
- Heavily modified engines

More advanced deep learning approaches can also be explored. Techniques such as **transfer learning**, residual networks, attention mechanisms, and ensemble models may provide higher accuracy than the current baseline CNN architecture.

Another useful extension would be **mobile application deployment**. A simple app could allow users to record engine sound, upload the sample, and receive possible fault predictions instantly.

The system may also be integrated with **smart garage environments**, where microphones installed in service bays automatically analyse vehicle sounds and assist technicians during inspection.

Additional future enhancements may include multilingual user interfaces, maintenance suggestions based on detected faults, cloud-based data storage, and predictive maintenance alerts.

Overall, the project has strong future potential and can be developed into a practical intelligent diagnostic tool for modern vehicles.

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Appendices

This section provides additional technical details, implementation support, and dataset information related to the proposed engine fault detection system.

The dataset used in this study consists of the AI Mechanic Dataset along with 11 additional real-world engine audio recordings collected from publicly available sources. All audio samples were standardized to a sampling rate of 22,050 Hz, converted to mono channel, and segmented into fixed 3-second clips. The dataset was categorized into five classes: Normal Engine, Combustion Fault, Mechanical Fault, Belt/Accessory Fault, and Background Noise.

To improve model robustness, augmentation techniques such as time shifting, Gaussian noise addition, and pitch variation were applied. These techniques helped simulate real-world variations and reduced overfitting during training.

The Convolutional Neural Network (CNN) model was trained using the following parameters: input size of 128 × 128 × 1, batch size of 32, 12 epochs, Adam optimizer, and Sparse Categorical Crossentropy loss function. A train-test split of 80:20 was used along with 5-fold cross-validation to ensure reliable performance evaluation.

A sample code snippet used for audio preprocessing and Mel-spectrogram generation is provided below:

```
import librosa
import numpy as np

# Load audio file
signal, sr = librosa.load("audio.wav", sr=22050)

# Generate Mel-spectrogram
mel_spec = librosa.feature.melspectrogram(y=signal, sr=sr, n_mels=128)
mel_spec_db = librosa.power_to_db(mel_spec, ref=np.max)

# Normalize
mel_spec_db = (mel_spec_db - mel_spec_db.min()) / (mel_spec_db.max() - mel_spec_db.min())
```

The complete source code, trained model details, and additional outputs are available in the GitHub repository:
https://github.com/harsimrat11/Engine_Fault_detection_using_CNN